

BANKING SERVICES PROPOSAL: SCORECARD FOR EVALUATING RETAIL CREDIT APPLICATIONS

HOMework TEAM 4

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Overview

The Commercial Banking Corporation's Department of *Revolving Lines of Credit* has requested the creation of a scorecard to be used for evaluating all retail credit applications. Our team thus developed a scorecard to meet this need and optimize profitability based on statistical significance. Based on our model, we recommend automatically approving customers above 493, manually reviewing those with a score between 463 to 492, and rejecting those below 463.

Methodology

The following section describes the format and preparation of the data used to produce our initial and final scorecards.

Data Used

Our team developed a scorecard using a dataset of previously accepted applicants and a dataset of previously rejected applicants.

The rejected applicants dataset contains information on applicants who were not approved. This dataset included 1,500 observations and 22 variables.

The accepted applicants dataset contains information on applicants who were approved. The dataset includes 3,000 observations and 24 variables. This dataset shared all of the same variables with the rejected data, with the addition of two variables: "good/bad" classifier as well as the oversample weights. The sample is balanced, with 1,500 "good" applicants (*i.e., those who have never been 90 days past due on payment*) and 1,500 "bad" applicants (*i.e., those who have been 90 days past due on payment at least once*). There are no indeterminate cases in this dataset. The original development sample had an actual bad rate of 3.23%, and over-sampling corrected this rate with verified weights.

Data Preprocessing

To prepare for fitting our initial scorecard, we split the accepted applicants data using the Bank's suggested 70:30 train-test split. We binned variables and performed variable selection using Information Values (IVs), which ranks variables on the basis of their importance. We retained only those variables with an IV between 0.1 and 1.0 to ensure meaningful predictive strength.

Analysis

This section describes the development and evaluation of the initial credit scorecard, the implementation of reject inference, and the performance of the final scorecard.

Initial Scorecard Development and Evaluation

Using the accepted applicant training data, we developed an initial credit scoring model by fitting a logistic regression to the Weight of Evidence (WoE) transformed variables. The WoE transformation standardizes predictors and ensures a more linear relationship with the log-odds of default, making them suitable for logistic regression modeling. We used this model to construct the initial scorecard.

Our initial scorecard had an area under the ROC curve (AUC) of 0.72, indicating a strong ability to distinguish “good” applicants from “bad” applicants. This scorecard had a K-S statistic of 0.33 with an optimal probability cutoff of 0.03, which classifies any applicant with a predicted default probability above 0.03 as a “bad” applicant.

Reject Inference

We implemented reject inference to align with FDIC requirements and create a scorecard that can generalize to the entire credit population. Using our initial scorecard, we calculated predicted probabilities of default for the dataset of rejected applicants. We then classified these rejected applicants as “good” or “bad” using the same 0.03 cutoff from our initial scorecard evaluation.

We combined the 1,500 rejected applicants with the 3000 accepted applicants to remove the bias generated by excluding rejects. After combining both applicant pools, we obtained a split of 67% accepted applicants and 33% rejected applicants. This split differs from the Bank’s current acceptance rate of 75%. To align our distribution with this rate, we reweighted accepted applicants while maintaining the Bank’s current default rate of 3.23%. This approach ensured that the impact of the accepted applicants in our sample would generalize to the 75% of future applicants that the Bank will accept.

The same processes of train-test splitting, variable binning, and fitting the credit scoring model were performed to produce the final scorecard.

Results & Recommendations

In this section, we will discuss the process for determining the final cutoff.

Final Scorecard Results

After incorporating reject inference, the final scorecard achieved an AUC of 0.73, demonstrating strong discriminatory power between good and bad applicants under the expanded modeling framework.

Figure 1 visualizes a decile plot that bins predicted scores from the test dataset into ten equal pieces.

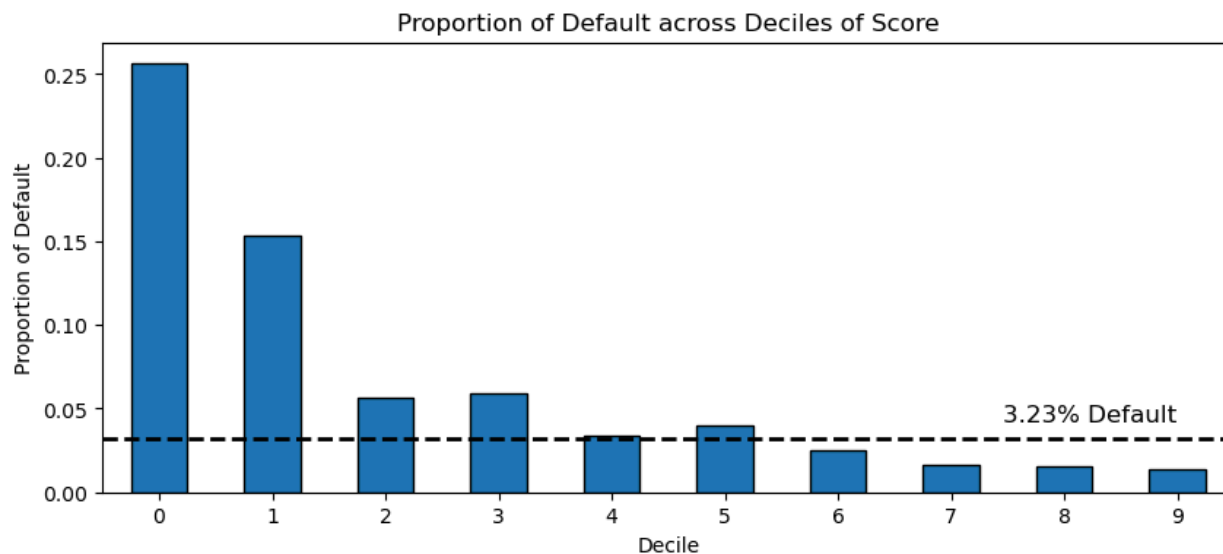


Figure 1: Proportion of Default across Deciles of Score

As shown in Figure 1, the proportion of defaults generally decreases. This trend indicates that our scorecard correctly assigned lower scores to applicants who had higher default probabilities, and higher scores to applicants who had lower default probabilities.

Scorecard Cutoff Results

To determine the optimal cutoff, we evaluated how the credit score cutoff impacts the Bank's default and acceptance rate. A line graph depicting this relationship is shown in Figure 2.

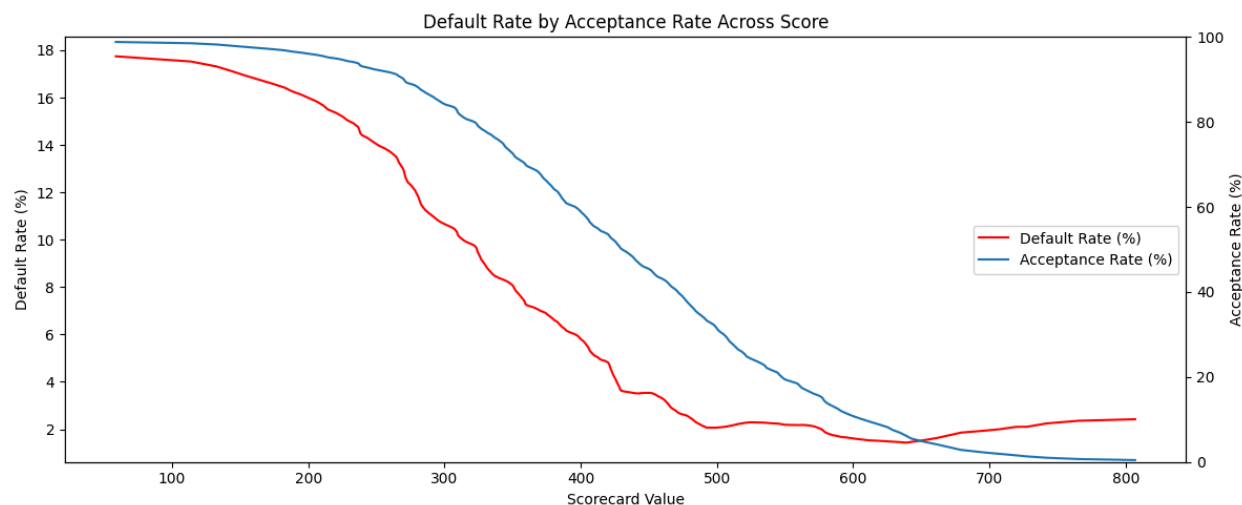


Figure 2: Default Rate by Acceptance Rate Across Score

From Figure 2, our team observed that the default rate decreases steadily as the credit score cutoff increases, which indicates improved borrower quality. Similarly, the acceptance rate also decreases as the credit score cutoff increases. At low score thresholds, almost all applicants are approved, but default rates peak. The converse is seen at higher score thresholds. While Figure 2 identifies the score range where the optimal cutoff lies, it does not take into account the Bank's profit margins. Figure 3 shows the relationship between the Bank's profit and acceptance rate.



Figure 3: Profit by Acceptance Rate Across Score

The acceptance rate visualized in Figure 3 shows the same pattern from Figure 2. However, the profit curve is mostly increasing, peaking at a credit score of 493. This peak represents the point at which the Bank's profit is maximized. Scores beyond this peak slowly begin to decrease the Bank's profit.

Therefore, for evaluating all retail credit applications, we recommend that the Bank automatically accept applicants with credit scores above 493 to optimize profitability. Additionally, we recommend that the Bank reject applicants with scores below 463 to maintain stability of the current default rate. Scores that fall within this range should be reviewed manually to optimize profitability.

Conclusion

This scorecard maintains the Bank's current acceptance rate of 75% while maximizing profit. Our final model is not only more representative of the applicant population, which improves accuracy and reduces bias, but also seems to discriminate well with an AUC of 0.73 and a relatively decreasing decile chart.

Overall, based on the Bank's revenue, cost requirements, and the tradeoff between acceptance and default rates, we recommend a range where customers above 493 are automatically accepted while those below 463 are automatically rejected. For scores between 463 to 492, we recommend manually reviewing those applications to better optimize profitability.